

Guidelines and best practices for the scientific use of global iNaturalist data

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Abstract

iNaturalist data are increasingly used in scientific research, yet researchers often lack clear guidance on how to access, handle, and properly attribute these data. Here, we provide a practical guide for the effective use of iNaturalist data in biodiversity research. We begin with an overview of key data concepts, including taxonomy, geoprivacy, location accuracy, and data availability. We then outline methods for acquiring biodiversity observation data, data summaries, and media, and provide example code in R and Python. We discuss key analytical considerations, including observer bias, the presence-only nature of the data, and appropriate modeling approaches. Finally, we highlight best practices for ensuring reproducibility through proper attribution, versioning, and transparent reporting; and we identify emerging research frontiers. This guide aims to support the responsible and effective use of iNaturalist data to further accelerate biodiversity research.

Keywords iNaturalist, citizen science, recommendations, community science, participatory science

Rationale and scope

The majority of contemporary global biodiversity data are now contributed by millions of participants through participatory science (often referred to as citizen or community science) platforms, with at least 80% of recent contributions to the Global Biodiversity Informa-

tion Facility (GBIF) generated through these contributor communities (Callaghan et al. 2021). Among these communities, contributors to iNaturalist collectively generate the broadest geographic and taxonomic coverage of openly available biodiversity observations (Di Cecco et al. 2021), with the volume of contributed observations grow-

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ing at a mean rate of 50% per year over the last decade. The volume of scientific research leveraging data from iNaturalist is also increasing at rapid rates, with a tenfold increase in publications using iNaturalist data in the last 5 years (Mason et al. 2025). This growth aligns with an increase in policy-usage of iNaturalist data (e.g., environmental impact statements; Callaghan et al. 2025). Despite this rapid uptake, researchers do not always have training in the best practices for understanding and handling iNaturalist data, knowledge of how to efficiently and appropriately access and attribute data, and guidance in how to work with the data most effectively. Because every biodiversity data point originates from the efforts of individual observers and identifiers, responsible scientific use requires guidance on the understanding of the data generation process, as well as the collaborative processes through which those data are generated and maintained.

As with most participatory science datasets, iNaturalist data are uploaded by many different observers from different parts of the world. Attributes inherent to participatory science (e.g., variation in observation effort, identification accuracy, and metadata completeness) can affect and inform the resulting utility of the data. This includes variation in the spatial and temporal accuracy of observations (Feldman et al. 2021), the quality and completeness of image metadata (Pernat et al. 2024a), and the spatial coverage of observations across landscapes (Geurts et al. 2023a). Because the data are used across disciplines—ranging from ecology to computer science to education—there are heterogeneous and diverse uses of, and caveats to, the data depending on the research questions. There is therefore a need to summarize the current best practices and guidelines for accessing the necessary data, understanding its structure, and using the data in practice.

We provide the first comprehensive methodological guide for effectively using global iNaturalist data in scientific research (figure 1). This effort is inspired by similar guides to using eBird data in research (Johnston et al. 2021, Strimas-Mackey et al. 2023). We systematically explain how iNaturalist observations are generated, curated, structured, accessed, and analyzed, with an emphasis on the implications of these processes for biodiversity research. Specifically, we provide an overview of iNaturalist data and foundational concepts (box 1), describe the critical data components and metadata required to interpret observations, outline the major pathways for accessing data (GBIF, the Export Tool, and the API), discuss analytical considerations and best practices for working with this complex, dynamic dataset, and provide guidance on attribution and reproducibility. We additionally provide code examples in R and Python for some of the above, as well as provide perspectives on the future of iNaturalist data usage. Although our focus is on biodiversity research, we believe these guidelines will be pertinent to many different research disciplines (e.g., social sciences, computer science) and anticipate that these guidelines will facilitate the further use of iNaturalist data for scientific research and policy implementation.

Box 1: Foundational concepts for using iNaturalist data.

iNaturalist is a global platform where users can upload photos or audio recordings to the iNaturalist platform through a smartphone or the website as evidence of an organism at a certain location and time. As an observation is uploaded, iNaturalist's computer vision model is available to suggest species-level identifications based on the submitted media and geographic context. The observing user can then

review these candidate suggestions as they develop an initial identification. The community then helps identify the observation to the finest possible taxonomic resolution (Callaghan et al. 2022, Campbell et al. 2023). Consequently, each observation represents the product of an ongoing collaborative process involving observers, identifiers, curators, and other contributors. Although most identifications are made by experienced users (Campbell et al. 2023), there are no requirements to become an identifier, and no extra weight is given to experts.

A key piece of information in iNaturalist data is the “quality grade” of a given record: “Research Grade,” “Needs ID,” or “Casual.” Broadly, Research Grade status is attained when an observation meets the site's metadata quality criteria, and has two or more suggested identifications, more than two-thirds of which agree at a species level or a taxonomic level finer than family when further progress in identification is deemed not possible (i.e., 2/2, 3/3, 3/4, etc.). Observations classified as “Needs ID” meet the metadata requirements for Research Grade but have not yet reached community consensus on their taxonomic identification. Because community identifications accumulate over time, recently uploaded observations are less likely to have reached identification consensus than older observations. Researchers using very recent data should therefore recognize that some observations may still be awaiting community review. Note that Research Grade does not guarantee a correct identification, and researchers should take care to verify observations themselves as appropriate. By contrast, “Casual” observations are those that lack required information (e.g., photo or audio evidence, location, or date), are cultivated, captive, or otherwise not “wild,” cannot be identified to a finer taxonomic level than family, and are marked as such through community annotation. See glossary for more details on these key “observation quality grades.”

The iNaturalist infrastructure provides mechanisms for organizing these different types of observations into structured collections. Most notably, users can create “Projects” (see Glossary), which other users may join or contribute to. Projects allow observations to be grouped, curated, and enriched around a shared research question, geographic region, taxonomic focus, or community initiative. Projects allow for observations to be systematically organized, filtered, and augmented with additional metadata (e.g., through observation fields), enabling coordinated data collection and curation within an otherwise open and dynamic dataset. Observations can be included in an unlimited number of projects.

Another important component of the data submitted to iNaturalist is the licensing. Three types of content can be licensed separately on iNaturalist: (1) observation data (including metadata such as date, geographic location, species identification, etc.), (2) images, and (3) audio recordings (iNatHelp 2024). By default, when creating a new account, all three of these groups are set to CC BY-NC (Creative Commons: Attribution-NonCommercial), meaning that the content can be used with attribution for non-commercial purposes. Other licensing options that users can choose are CC0 (No Copyright), CC BY (Attribution), CC BY-NC-SA (Attribution-NonCommercial-ShareAlike), CC BY-NC-ND (Attribution-NonCommercial-NoDerivs), CC BY-ND (Attribution-NoDerivs), CC BY-SA (Attribution-ShareAlike), and no license (all rights reserved). Licenses are set independently for observations, images, and audio recordings, and can differ within a single observation (figure 2). These licensing settings govern whether and to what extent researchers may legally reuse iNaturalist content. As of July 2026, approximately 23% of iNaturalist observations are not available under an open observation license and therefore can-

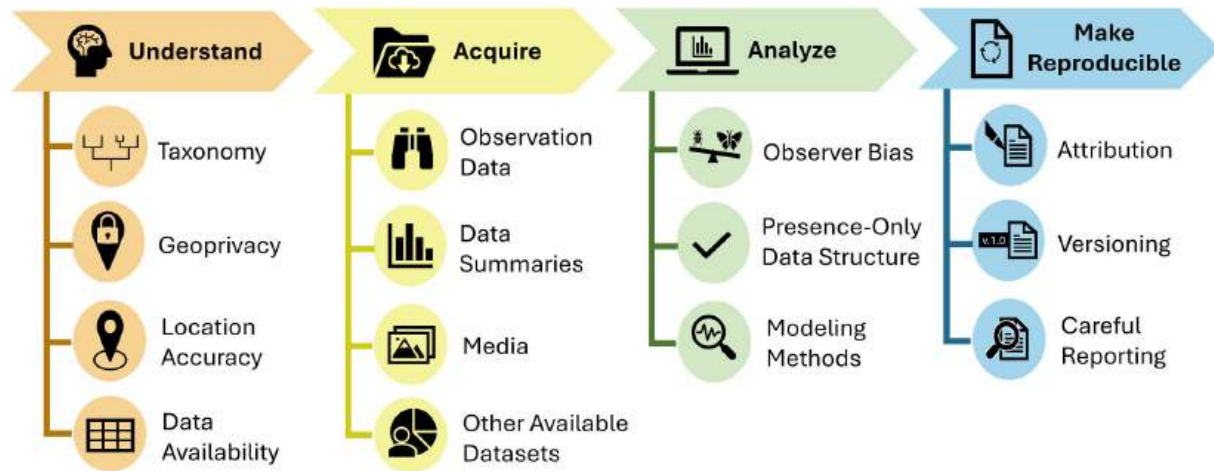


Figure 1. Using iNaturalist data effectively in scientific research requires understanding at multiple levels how to interpret the data, acquire the data, analyze the data, and ensure reproducibility.

not be redistributed through GBIF, highlighting how licensing choices substantially influence the availability of observations for scientific research. If no license is set, then the data are considered “all rights reserved.” iNaturalist generates a weekly export of observations eligible for inclusion in GBIF (Heberling et al. 2021) through which researchers can access biodiversity data. Exported observations include only Research Grade observations that are licensed as CC0, CC BY, or CC BY-NC (default option).

A final important concept to understand about iNaturalist data is that it is dynamic. Identifications, taxonomic definitions, metadata, and even licensing settings can change over time as community members contribute new information or users update their content. As a result, datasets retrieved for scientific research on different dates may differ, and reproducible workflows should explicitly document download dates, filters, and observation identifiers.

Critical data components and metadata for working with iNaturalist data

This section details critical data components and metadata from iNaturalist, including geoprivacy settings, obscuration, data accuracy, and the types of data available on the platform. For components of taxonomic information, see box 2.

Geoprivacy, Obscuration, and Data Accuracy

iNaturalist observations vary in their spatial precision due to geoprivacy settings, coordinate obscuration, user-defined positional uncertainty, and differences in GPS precision, with measures of coordinate uncertainty available through both GBIF and the iNaturalist API or the iNaturalist Export Tool. By default, observations uploaded to iNaturalist have their geoprivacy set to “open,” meaning that the user permits the data to be publicly shared, and other users can freely see the coordinates and date/time of these records. However, some iNaturalist observations have their geoprivacy changed to “obscured,” which can occur in two different ways. First, users may choose to manually obscure any of their records (referred to as “geoprivacy”) at any time for privacy or other personal reasons.

Second, some observations are automatically obscured by iNaturalist (referred to as “taxon geoprivacy”; see Supplemental Text S1 for more information).

When an observation is obscured, iNaturalist hides the original latitude and longitude, replaces the publicly displayed coordinates with a randomly assigned location within a $0.2^\circ \times 0.2^\circ$ grid cell, and sets the public coordinate uncertainty (positional accuracy) to the length of that grid cell’s diagonal (~ 31 km at the equator; decreasing in size toward the poles; Supplemental Text S1–S2). In GBIF, obscured observations are not explicitly flagged; instead, the obscured coordinates and their associated coordinate uncertainty are reported. Information on whether an observation has been obscured is only available from iNaturalist through the geoprivacy metadata field, along with the obscured coordinates and coordinate uncertainty. “Private” observations, a third type of geoprivacy, share no geographic information publicly, and these observations are not shared with GBIF. Users may share obscured or private coordinates by trusting a project or individual user (iNatHelp 2023). Network nodes (i.e., regional or national partners that operate localized versions of the iNaturalist platform; iNaturalist 2026a) also get access to true coordinates of observations made by users who have chosen to affiliate with them. If granted access to these hidden coordinates, take special care to avoid sharing this sensitive data in publications or data releases.

Users may also upload observations with intentionally coarse positional accuracy for the coordinates (Supplemental Text S1). This may occur because users do not remember the precise location at which they made the observation, or as an alternative method to obscure the location.

Data available on iNaturalist

The most commonly used data from iNaturalist are date and time of an observation, consensus taxonomic identification (hereafter Community Taxon), and geographic location (Mason et al. 2025). Spatiotemporal data from iNaturalist are used by researchers for a myriad of reasons, including to describe species ranges and distributions (Cull 2022, Wolfe et al. 2025), assess biodiversity trends (Beninde et al. 2023), compile species lists (Wenk et al. 2024), and more.

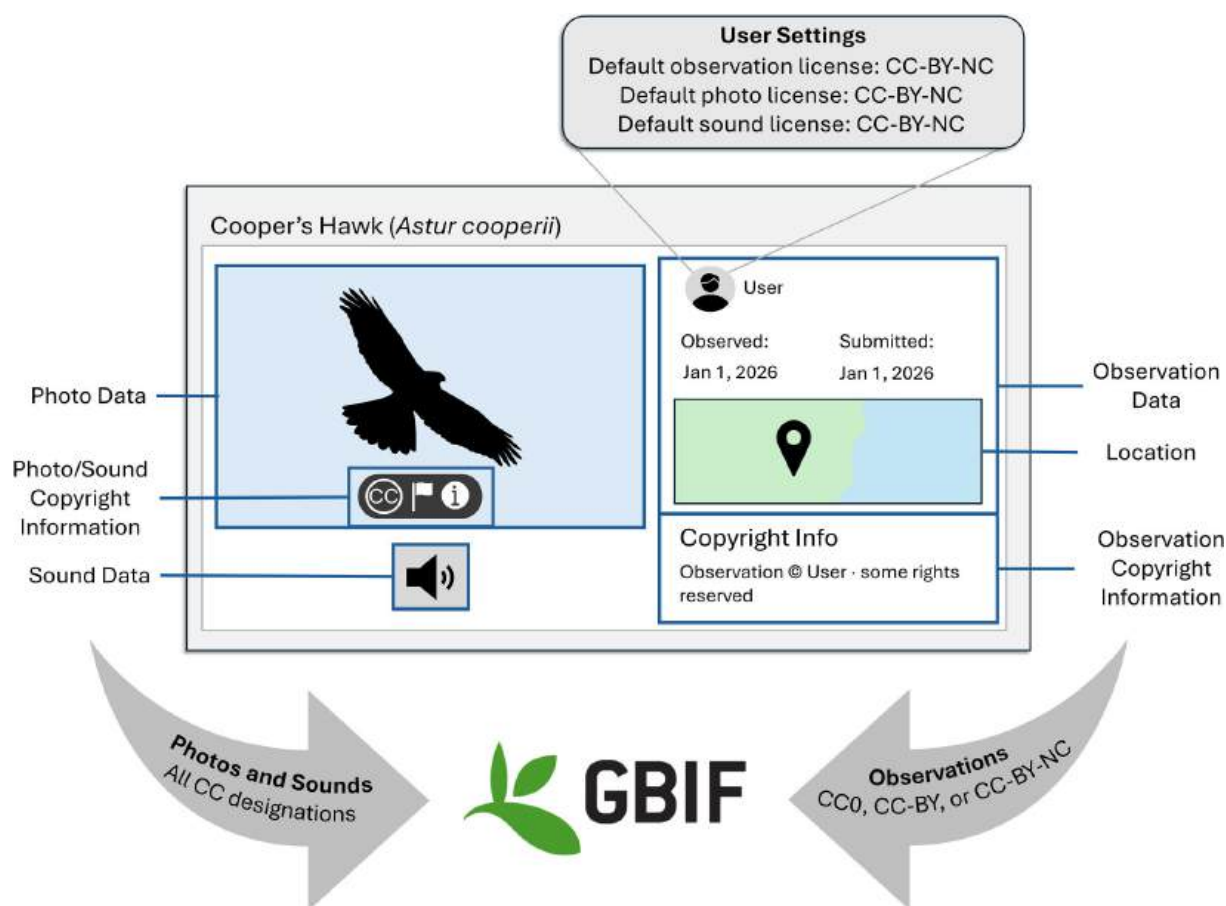


Figure 2. Schematic of an iNaturalist observation from the iNaturalist website illustrating the three independently copyrighted data sources associated with iNaturalist records: photo data, sound data, and observation data. The diagram also shows where copyright and licensing information can be found for each data type. Photos and sounds with any Creative Commons (CC) designation are shared with GBIF for scientific use, whereas observations must be designated as CC0, CC-BY, or CC-BY-NC (the default) to be shared with GBIF. Researchers are responsible for ensuring that data are used in accordance with the applicable licenses.

Box 2: iNaturalist has a dynamic taxonomic backbone.

iNaturalist's taxonomy is highly dynamic, with changes—including the addition, synonymization, and splitting of taxa—implemented daily by site curators to reflect changes in both primary (the scientific literature) and secondary (taxonomic database) sources. Broadly, curators attempt to align iNaturalist taxonomy to global taxonomic authorities for taxa where these exist, such as Plants of the World Online (Royal Botanic Gardens Kew 2026) for vascular plants or The World Spider Catalog (Natural History Museum Bern 2026) for spiders; a [list of these external authorities](#) is available for curators on the site (see [How Taxon Changes Work](#) for complete details on how this affects iNaturalist data). Although following designated external authorities as closely as possible is strongly encouraged, curators can “deviate” from these authorities where appropriate to follow differing regional treatments. Regional taxonomic authorities and the primary literature are referenced for taxa where global taxonomic authorities do not exist, such as fungi or many terrestrial invertebrate groups.

iNaturalist does not maintain or publish historical versions of its taxonomic tree, so if it is to be referenced alongside an iNaturalist data download, both should be obtained at the same time. The latest version of the tree can be downloaded through

the [iNaturalist Taxonomy DarwinCore Archive](#), which is generated monthly. It is important to note that, because iNaturalist is an observational database rather than a nomenclatural database, it is not intended to represent an exhaustive source of all names for all taxa. The iNaturalist taxonomic tree differs from the one used by GBIF. When iNaturalist shares data to GBIF, GBIF uses its own [taxon match service](#) to align taxa with its own taxonomy. Until June 2026, this was [GBIF's taxonomic backbone](#), but it now matches to both the legacy backbone and also to the [Catalogue of Life XR \(COL XR\)](#), the new taxonomic backbone for GBIF (Bánki et al. 2026). When name mismatches occur, GBIF may use fuzzy matching or match to a coarser taxon. The legacy GBIF Backbone Taxonomy was last updated in 2023, and is retained for backward compatibility. The new COL XR Backbone is updated monthly. Both the legacy backbone and the COL XR backbone can be downloaded from <https://www.checklistbank.org/>.

Species lists, summarized species count data, and range maps can be a useful alternative to downloading a full dataset when fine-scale detail is not required. These include summarized species counts for specified parameters (e.g., place, taxon, user) and summarized counts by time (e.g., week of the year, month, year). Furthermore, iNaturalist provides over 100,000 range maps that have been modeled from iNat-

uralist data (see more [here](#), including information on how the maps were made; [Supplemental Text S12](#); iNaturalist 2026b).

Media submitted to iNaturalist are also a rich data source that is increasingly used by researchers (Mason et al. 2025). To extract useful data from images and audio files, such as behavior or morphology, researchers may manually review media and record relevant data, or they may use advanced computational techniques such as supervised classification or computer vision models (Perez-Udell et al. 2023). Audio data are less commonly used in the literature, but examples include studying anuran breeding phenology (Forti et al. 2022) and testing the ability of convolutional neural networks to accurately identify species based on their calls (Wu et al. 2022). Additional data types, such as identification history and user activity trends, are available for special use cases (see [Supplemental Text S1](#) for examples).

Additional data fields that can be added to iNaturalist observations include annotations, observation fields, and tags (see table 1), which can be accessed through the iNaturalist Export Tool or API but are not available on GBIF. These fields are optional and can be contributed by the original observer or, in the case of annotations and observation fields, by the broader community—unless the observer has restricted who can add fields to their observations. Annotations are a limited set of fields with prescribed values that are decided by the iNaturalist administrators, with options including, but not limited to, “alive or dead,” “evidence of presence,” “life stage,” “sex,” “flowers and fruits,” and “leaves.” Some annotations are only available for certain taxa; for example, the flowers and fruits and leaves fields are only available for plants where those attributes are applicable. Although most records are not annotated, the number of annotated records is increasing each year and can provide valuable insights (figure 3). Annotations may be useful for studying phenology (e.g., timing of flowering) or providing human labels to train machine learning models (Dinnage et al. 2025), studying differing life stages (e.g., timing of hatchling emergence), or filtering the data (e.g., removing dead animal observations prior to examining phenology). Additional data are available from observation fields, which are often associated with specific projects that seek to gather additional metadata, or less commonly, tags, which are user-generated keywords that can be added to any observation by the observer (see [Supplemental Text S3](#) for more details on these fields).

Workflows, and their tradeoffs, for accessing iNaturalist data

Researchers should select a data-access workflow based on the data required for their research question while maximizing reproducibility whenever possible (figure 4). We recommend using the simplest workflow that provides the necessary information, with GBIF serving as the preferred access point whenever it contains the required data. Detailed tutorials for each workflow, including example queries and code, are provided in [Supplemental Text S11](#).

For most biodiversity studies, we recommend downloading iNaturalist data through GBIF (<https://www.gbif.org/>), which aggregates all Research Grade observations released under CC0, CC BY, or CC BY-NC licenses. In addition to providing standardized Darwin Core occurrence records, GBIF assigns a DOI to every downloaded dataset, creating a permanent, citable snapshot of the exact data used in an analysis ([Supplemental Text S4](#)). This greatly simplifies reproducibility and provides a transparent chain of attribution back to iNaturalist contributors. Even when only a relatively small number of observa-

tions are required, we generally recommend downloading through GBIF whenever the necessary records and metadata are available. For studies requiring only one or a few individual observations, directly citing the corresponding iNaturalist observation URL or observation ID is generally sufficient.

When data unavailable through GBIF are required (e.g., Needs ID observations, observation fields, collection-project information, trusted coordinates, observations that are not licensed under CC0, CC BY, or CC BY-NC, or complete identification histories), we recommend using the iNaturalist Export Tool ([Supplemental Text S5–S6](#)). This web-based interface supports more complex filtering and provides access to additional metadata not included in GBIF. However, because Export Tool downloads do not generate versioned datasets or DOIs, researchers should carefully document download dates, search parameters, and archive the final dataset used for analysis.

The iNaturalist API should generally be reserved for situations where the required filters or returned data are unavailable through either GBIF or the Export Tool. Examples include specialized filters such as observations updated since a specified date, observations lacking taxonomic identifications, or excluding certain users from observation sets, as well as programmatic access to annotations and other nested metadata. This method provides the most granular control over both search filters and returned fields. For a full list of parameters, refer to the [iNaturalist API](#) documentation (iNaturalist 2026c; see [Supplemental Text S7](#) on how to modify request URLs for the API). Additionally, API calls by default return data in JSON format, which allows complex fields—such as media, annotations, and identifications—to be represented as nested data structures containing all relevant items and their associated metadata. Although the API offers the greatest flexibility, it also requires the greatest level of technical expertise and places the burden of reproducibility on the researcher. Accordingly, API users should archive derived datasets and report all endpoints, parameters, and access dates. Because the API is considerably more complex than the other access pathways, we provide detailed guidance, annotated example workflows in R and Python, and additional API resources in [Supplemental Texts S7–S10](#).

Regardless of the access route used, researchers should clearly document how data were obtained, report download dates and filters, and archive derived datasets where appropriate. These recommendations are expanded further in the Attribution and reproducibility section.

Best practices for analyzing iNaturalist data

iNaturalist is a large unstructured biodiversity dataset, created by observers who choose when, where, and what to record (Bowler et al. 2022). iNaturalist data are powerful for documenting spatiotemporal patterns of species occurrence, yet because the data are unstructured and heterogeneous across space, time, taxa, and observers, many of the underlying sources of heterogeneity are unreported (Geurts et al. 2023a). Scientific use requires understanding the data-generating process, matching analytical tools to the type of information the platform can provide, and applying careful filtering, modeling, and drawing appropriate inferences. In this section, we outline six overarching best practices, or topics of consideration, which range from technical decisions (e.g., filtering, accuracy thresholds) to conceptual decisions (e.g., scales of inference) to maximize the scientific use of iNaturalist data.

Table 1. Glossary of key terms used throughout this manuscript.

Term	Definition	Notes/implications for research use
Participatory Science	Engagement of the general public in the scientific endeavor. This approach is also referred to as <i>citizen science</i> or <i>community science</i> , although <i>participatory science</i> has become increasingly common because it is viewed as more inclusive. Additionally, <i>community science</i> may refer specifically to scientific efforts that are led by communities	Although we use the term participatory science throughout this paper, the scientific literature also uses the terms citizen science or community science
Observation	A record submitted to iNaturalist. Records that include media (photo/audio) and associated metadata (date, location) are deemed “verifiable”	Observations do not imply systematic sampling; absence of an observation of a species $\neq$ true absence of a species
Computer Vision	A machine-learning model that generates initial species identification suggestions from the user’s uploaded photos	Although initial identifications from this model can increase the likelihood that an observation reaches Research Grade, they should not be relied upon until they receive community support
Identification (ID)	A taxonomic label suggested by a user or computer vision for an observation. Multiple IDs can be attached to a single observation	Each ID is weighted equally, regardless of who suggested it
Community Taxon	The taxonomic consensus resulting from the community identification process based on a $>2/3$ majority agreement rule	IDs can change over time as more users review; the <i>community taxon</i> reflects consensus at a given point in time
Research Grade	A status assigned to observations that meet (1) verifiable metadata (date, location, media), (2) community taxon finer than family, and (3) $\geq 2/3$ ID agreement	Only Research Grade observations with appropriate licenses are exported to GBIF
Needs ID	Observations that are verifiable but have not yet reached community consensus at a fine taxonomic level	Often valuable in data-limited taxa; may include high-quality records missing expert review or only identified by a single expert
Casual	Observations lacking key metadata (e.g., no media or location), depicting captive/cultivated organisms, or having low-quality metadata (e.g., date is not accurate)	Usually excluded from biodiversity analyses unless the research question requires them
License	Usage rights chosen by observers for their observation data, images, and audio files (e.g., CC0, CC BY, CC BY-NC)	License type determines how data can be legally downloaded, shared, and exported to GBIF
Geoprivacy	Setting determining how precisely location data are displayed: <i>open</i> , <i>obscured</i> , or <i>private</i>	Obscured locations are randomized within a $0.2^\circ \times 0.2^\circ$ cell; private observations hide locations entirely and are not exported to GBIF
Obscured Observation	An observation whose coordinates are intentionally randomized to protect sensitive species or user privacy	Suitable for coarse-scale analyses; do not use for fine-scale habitat inference without buffers or without obtaining true coordinates
GBIF Export	The subset of iNaturalist observations and metadata shared automatically with GBIF under compatible licensing	Best for reproducibility because GBIF provides a DOI reference for citation
iNaturalist’s Export Tool	A web-based tool on iNaturalist that allows users to search for and download observation data	This method offers more filtering options and exports more data than GBIF, but queries are limited to 200,000 observations at a time, and DOI references are not provided
API Access	A programmatic method of requesting data directly from iNaturalist endpoints	Allows for full control and access to <i>all</i> metadata, but requires rate-limit awareness and documentation of query parameters
iNaturalist Open Data	Dataset of iNaturalist images, updated in real time, and their associated metadata, updated monthly. Hosted on Amazon Web Services	Photo URLs formatted as s3://inaturalist-open-data/photos or https://inaturalist-open-data.s3.amazonaws.com/photos are hosted on Amazon Web Services and can be downloaded without placing any load on the iNaturalist servers
Sampling Effort	The latent and uneven distribution of where, when, and what observers choose to record	Should be considered, and where relevant, modeled or otherwise accounted for to avoid misinterpreting sampling bias as ecological signal
Presence-only Data	Data describing where a species has been observed, without information on where it was searched for but <i>not</i> found	iNaturalist data are inherently presence-only; absences must be inferred statistically, never assumed
Reporting Rate	The fraction of all observations in a given unit that belong to a particular species	May approximate relative abundance if sampling effort is adequately modeled
Taxonomic Backbone	The taxonomic reference system used by iNaturalist to organize and reconcile species names	Changes over time; analysts should record the <i>date</i> when data were downloaded

We further refer the reader to this link with further terms relevant for iNaturalist: <https://www.inaturalist.org/terminology>.

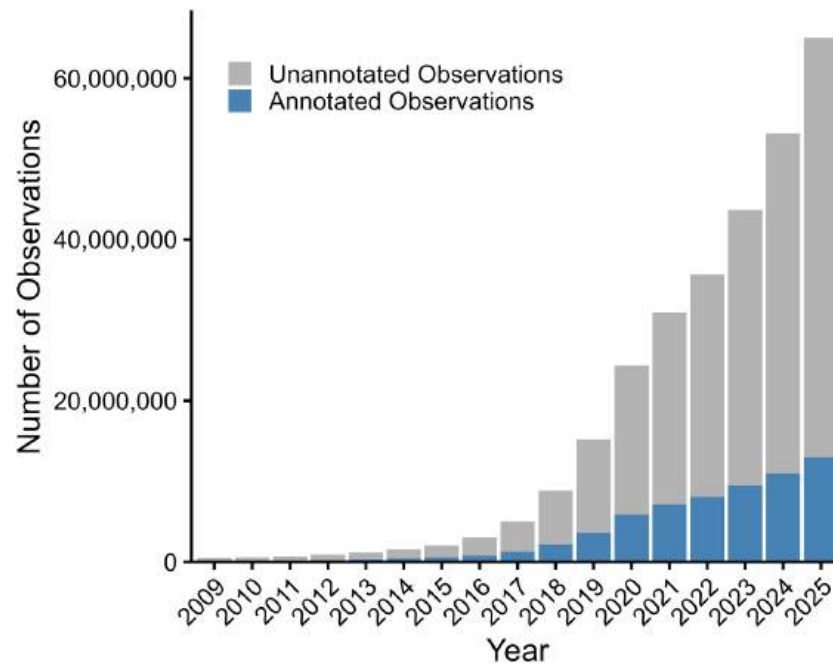


Figure 3. Number of observations with at least one annotation versus unannotated iNaturalist observations by year. Over 64 million observations made between 2009 and 2025 have at least one annotation.

Evaluate identification accuracy as necessary

iNaturalist species IDs are not perfectly accurate (e.g., McMullin and Allen 2022, Mesaglio et al. 2025). Even for easy-to-identify species, a small fraction of Research Grade observations may be misidentified, especially those based on indirect evidence (e.g., scat, footprints) rather than the organism itself. How much one must consider misidentifications should scale with the inferences being drawn. For example, if iNaturalist data are used for the detection of a threatened taxon or documenting novel range expansions, greater scrutiny should be placed on confirming the accuracy of the relevant observations. In contrast, a species distribution model fitted with tens of thousands of observations will not be sensitive to a handful of misidentified records (Graham et al. 2008), although sensitivity to false positives can vary widely depending on the statistical model (see, e.g., Clare et al. 2021, Van Eupen et al. 2021). In some cases, even using non-Research Grade data with some quality controls can provide useful information about biodiversity (Hochmair et al. 2020, Mesaglio et al. 2025). Because iNaturalist's identifications are dynamic and updated continuously, analysts should explicitly consider the potential for identification uncertainties and ensure that filtering choices (e.g., species-level versus genus-level identifications) align with the biological question.

Do not treat non-detections as absences

iNaturalist records are an extremely convenient source of information on species' presence. They are less suited, however, to establish the absence of a species. In areas with many iNaturalist records, a lack of observations of a common and readily identified species may strongly correspond to the absence of that species given that similar species are commonly reported in the area, but in areas of low overall iNaturalist effort or taxonomically specific iNaturalist effort, even common species may go unreported. Even the areas most saturated

with iNaturalist data do not document all species, and many that are low density, elusive, or especially diverse may simply not be detected. Conflating lack of iNaturalist data with absence is a dangerous but common mistake; in a review of iNaturalist data used in environmental impact reports, Callaghan and colleagues (2025) identified 28 cases where the lack of observations of a certain species was used to establish the absence of that species. We recommend against the use of iNaturalist data to provide evidence of species absence without the use of statistical analyses or supplemental professional sampling.

Apply question-specific filtering and quality control

Data filtering must match the research objective. When assessing or analyzing location data of iNaturalist observations, careful attention must be paid to whether coordinates have been obscured; otherwise, spurious conclusions may be drawn based on the randomized rather than the true coordinates of each record. For some questions, only Research Grade observations may be appropriate; for others, Needs ID records may contain valuable information, especially in data-limited taxa. In some cases, non-Research Grade observations do not have significantly worse taxonomic accuracy than Research Grade observations (Hochmair et al. 2020, Mesaglio et al. 2025). Similarly, geographic accuracy, date completeness, time stamps, annotations, and media requirements should be chosen intentionally. Filtering decisions should be explicitly justified in the context of the biological question, rather than applied as generic defaults. Choices about which fields to require, which taxonomic ranks to include, or how to treat incomplete meta-data directly shape the ecological inferences that can be drawn.

Match data treatment to the scale of inference

Across methodologies and taxa, analyses of iNaturalist data at very fine spatial scales can be especially challenging. Firstly, the geolocations associated with many iNaturalist observations have a level

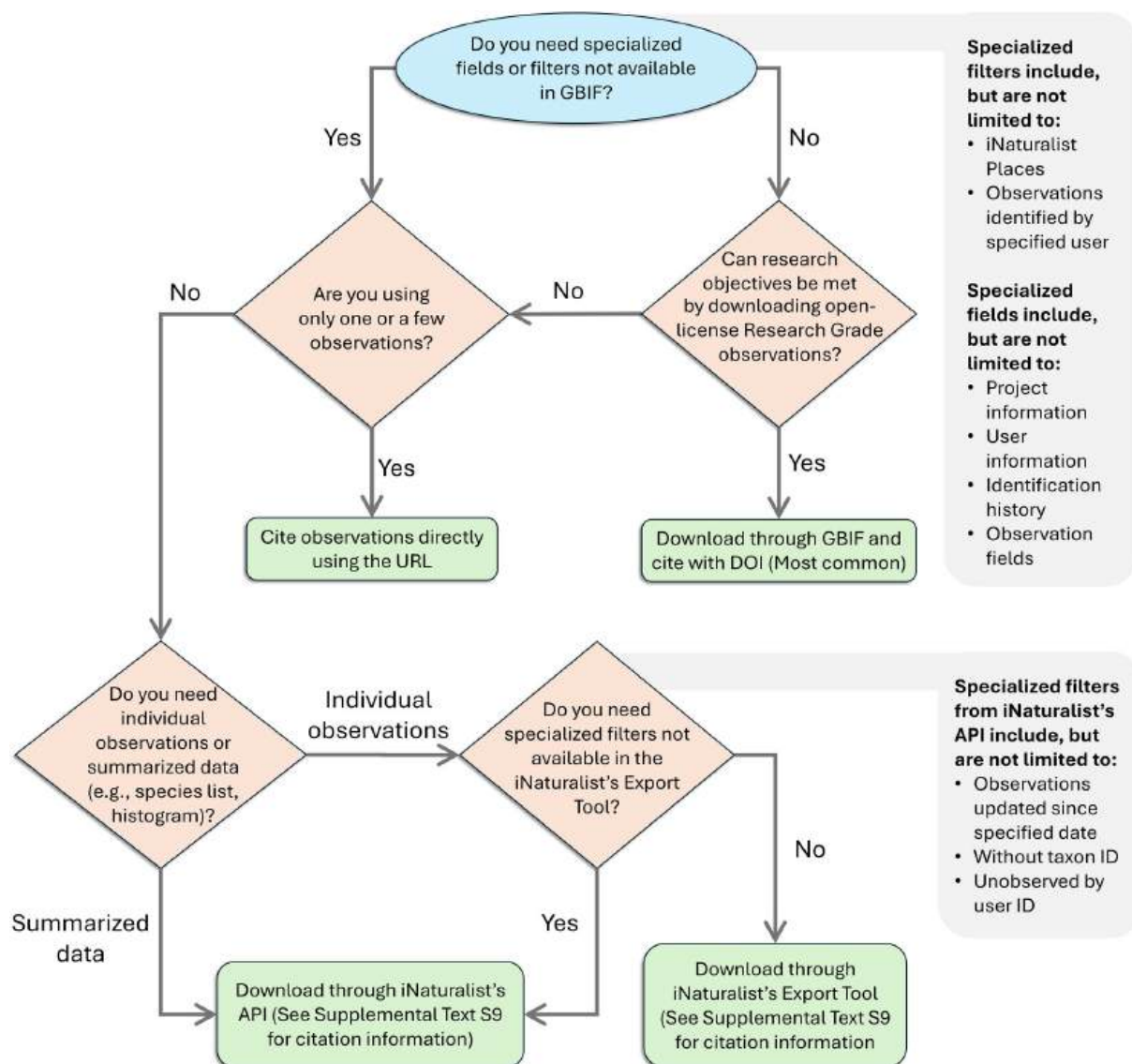


Figure 4. Flow chart to determine the best method of downloading iNaturalist observation data.

of uncertainty that is not conducive to very fine-scale applications (~10% of all Research Grade observations have a coordinate uncertainty of more than 500 m). Secondly, spatial variation in survey effort (where observers are not looking) becomes extreme at fine scales. Consequently, special care should be taken if modeling iNaturalist data at fine spatial scales (e.g., <1 km). The appropriate resolution will depend on the research question, taxon, sampling intensity, and treatment of positional uncertainty. Further, obscured records require special attention when considering scale of analyses, as obscuration replaces true coordinates with randomized coordinates within a $0.2^\circ \times 0.2^\circ$ grid cell. Researchers should determine whether these randomized coordinates are compatible with the resolution of their study, and when they are not, transparently report how these records were treated or excluded. Finally, if users filter observations by a specified threshold of positional uncertainty, they must also decide how to treat “NA” coordinate uncertainty values, which indicate missing information rather than necessarily high positional uncertainty.

The appropriate approach for dealing with record-level spatial uncertainty depends on the research objective. For analyses at coarse spatial scales, such as summaries of species occurrences across large regions (e.g., a state, province, county, etc.), coarse-grain data may still be useful at higher levels of uncertainty. However, for fine-scale analyses—for example, determining habitat associations or occurrences in small urban parks—obscured locations and locations with high uncertainty require careful treatment. In many cases, especially at fine spatial resolutions, records with obscured coordinates should not be used unless true coordinates can be assessed as they distort the basic information and analysis results (Koo et al. 2025). Typical approaches include filtering observations by a required level of location certainty, but in some cases, it may be reasonable to instead add buffers equal to the uncertainty radius around points and assess whether they overlap the area of interest. Analysts should be familiar with how positional uncertainty is determined in iNaturalist (see Supplemental Text S1).

Recognize and account for the heterogeneity of sampling

Users of iNaturalist data must contend with the fact that observation patterns arise from the behavior, preferences, and constraints of human observers (Grady et al. 2026). iNaturalist sampling is unstructured: observers post at their own discretion and typically do not report any metadata describing their search effort or motivations. Consequently, iNaturalist data arise from a sampling process that is uneven in space, time, and taxonomy. Thankfully, researchers have done a great deal of work characterizing these hidden processes. At the simplest level, iNaturalist sampling effort is clustered in times and places of high human activity, for example, near human population centers and along roads and trails (Geurts et al. 2023a, 2023b). Most observations are diurnal, and observer activity peaks in the spring and summer (though there is substantial geographic variation in seasonal patterns; Di Cecco et al. 2021, Forti and Szabo 2025). Such heterogeneity in sampling effort is compounded by mass-scale organized participation events (e.g., City Nature Challenge, Moth Week, Great Southern Bioblitz), which can create pronounced spatial and temporal pulses in observation effort that may influence analyses if left unaccounted for (Di Cecco et al. 2021). Observation effort also varies across taxa. Observers tend to preferentially report larger, rarer, and more colorful species (Callaghan et al. 2021, Stoudt et al. 2022). Additionally, because verifiable iNaturalist observations require evidence of the organism through photo or sound recording, there is a bias toward species that are easier to document based on size, behavior, activity patterns, and/or volume of their call (Arazy and Milkson 2021, Di Cecco et al. 2021, Forti and Szabo 2025). Individual observers also specialize in particular taxa and repeatedly observe them, meaning that search effort may be spatially clustered (Di Cecco et al. 2021, Goldstein and Stoudt 2025). This structured heterogeneity matters because it shapes detection probabilities, can bias richness estimates, and complicates comparisons across sites or taxa. Any use of iNaturalist data should consider this uneven sampling effort and ensure that the scientific question is robust to it or that these patterns are accounted for statistically.

A variety of specific modeling approaches to account for heterogeneity in iNaturalist data have been demonstrated in the literature. Many researchers retrieve and visualize iNaturalist data simply to establish places where the species occurs with certainty, then use critical thinking and interpretation of auxiliary data to weigh the roles of sampling effort and species absence in causing spatial gaps in species detections (e.g., Hobson et al. 2017, Jachowski et al. 2021). This approach is reasonable, especially for hard-to-detect species where iNaturalist data are often the best available information on a species' range and for applications where statistical inference of spatial effects is not of interest. Researchers may also vary their spatial scale of inferences so that the influence of heterogeneity is minimized. For instance, researchers found that the occurrence of common species was better documented at the county level as opposed to the town level (Bertin and Grima 2026). Another common approach is to thin iNaturalist data to standardize effort before modeling occurrences as presence-only data under the assumption that effort is constant (e.g., Dimson et al. 2023, Hill et al. 2024). For example, an analyst may retain only one observation of the focal taxon (or focal taxonomic group) per 10-km spatial grid cell. This approach may be appropriate for some cases, but it breaks down when sampling effort varies dramatically at the scale of spatial control. Furthermore, emerging analytical approaches

are improving the ability to use presence-only data to more accurately estimate species distribution models (Valavi et al. 2022). Researchers may also account for missing data by utilizing covariates and knowledge of the system in statistical workflows (Bowler et al. 2025). Finally, some approaches generate non-detection data to represent heterogeneous sampling (see the "Estimating non-detections" section).

Make clear inferences in the context of how iNaturalist operates

Many observer tendencies are consistent *across* broad gradients—people often observe similar kinds of organisms in similar ways in urban, suburban, and rural settings—but the *intensity* of observation effort can vary dramatically along these same gradients (Di Cecco et al. 2021). This means that some questions (e.g., coarse-scale range limits or broad biogeographic patterns) are robust to uneven effort, whereas others (e.g., local abundance or fine-scale habitat preferences) are far more sensitive to it. For instance, research comparing reptile and amphibian occurrence data from iNaturalist with data from traditional surveys found that iNaturalist observations were useful for estimating species richness but were not sufficiently robust for analyzing community structure or diversity without statistical techniques to correct for sampling bias (Marsh et al. 2026). Similarly, researchers evaluating the use of iNaturalist to characterize a mass mortality event caused by avian influenza found that the platform could identify the species affected, as well as the timing and location of mortality events, but could not reliably quantify the magnitude or underlying mechanisms of mortality (Taylor et al. 2024). Researchers should therefore interpret results conservatively when appropriate, articulate how sampling processes may interact with their hypotheses, and transparently report the assumptions and uncertainties underlying their inferences.

Attribution and reproducibility in published research using iNaturalist data

One of the many strengths of iNaturalist data is that they are "living," meaning that almost any aspect, including their existence, could change between iterative downloads. This dynamism allows for the improvement of taxonomic identifications, addition of data quality annotations, and correction of errors, but it can also result in meaningful changes to existing iNaturalist data at any time. This might mean that observations are deleted; or their licensing, taxonomic identification, geographic associations, or other attributes changed. It also means that analyses are not always exactly replicable unless the originally used data are available. At present, articles that do not provide the versioned dataset used for analysis often lack sufficient information for the dataset to be reconstructed (Supplemental Text S13). To account for this, we suggest that at a minimum, researchers explicitly document how and when the data were accessed.

As discussed above in detail, GBIF-mediated downloads generate versioned, citable datasets that preserve the state of iNaturalist records at the time of download for at least 6 months and often longer (see Supplemental Text S8 for further details). In contrast, data obtained through the iNaturalist API or export tool require users to manage their own versioning, including retaining downloaded data locally and documenting all filters, endpoints, and access dates, which should be reported and end products archived. These key data man-

agement aspects are essential for scientific transparency and compliance with FAIR (Findable, Accessible, Interoperable, and Reusable) principles, which are increasingly recognized for their importance by journals, funders, and researchers in ecology and evolution research (Manzano and Julier 2021, Koivisto and Mäntylä 2024). Although FAIR principles should guide data, metadata, and other digital objects (e.g., code) created by researchers, community science datasets also require respect for contributor decisions about licensing and CARE principles for contributor-generated data (Jennings et al. 2023).

Below, we provide a concise checklist of information that should be included when publishing analyses using iNaturalist data. These recommendations are intended to improve reproducibility while recognizing that iNaturalist datasets are generated through the collective efforts of a global community of observers and identifiers whose contributions underpin biodiversity research. Following these steps will maximize reproducibility, ensure proper attribution to iNaturalist contributors, that is, observers and identifiers, and support the long-term integrity of biodiversity data workflows (see [Supplemental Text S14](#) for reporting templates).

1. Include the pathway to data access.

- (a) Report GBIF download DOI(s), including the exact query used.
- (b) Report iNaturalist API endpoints or Export Tool URLs used, including all parameters.
- (c) State whether multiple data sources were combined (e.g., GBIF + API).

2. Specify the exact filters applied.

- (a) Specify quality grades included (Research Grade, Needs ID, Casual).
- (b) Specify taxonomic, geographic, temporal, annotation, and observation field filters.
- (c) Report any user-level filters applied (e.g., exclusion lists, project-based queries).

3. Document download and access dates.

- (a) Report the date(s) of data retrieval.
- (b) Specify any “created on” query used to define a terminal inclusion date prior to download.

4. Report licensing and usage conditions.

- (a) Summarize the licenses attributed to observations and associated media.
- (b) Describe how licensing restrictions were enforced (e.g., exclusion of “all rights reserved” content).
- (c) Confirm compliance with all applicable license terms.

5. State taxonomic versioning details.

- (a) Report the date of taxonomic tree or backbone retrieval, if used.
- (b) Describe how taxonomic changes (splits, lumps, renames) were handled.
- (c) Specify whether iNaturalist or GBIF taxonomic backbones were used.

6. Include geoprivacy and coordinate resolution handling.

- (a) Describe how obscured and/or private observations were treated.

- (b) Specify thresholds used for coordinate uncertainty filtering.
- (c) Report any buffers or uncertainty-based exclusion rules applied.

7. Archive data in a permanent repository.

- (a) Specify where processed datasets are archived (e.g., Zenodo, GitHub, Dryad).
- (b) Archive all observation IDs used in analyses where permissible, noting that trusted coordinates cannot be re-shared and that archived datasets may therefore be incomplete.

Emerging research frontiers for iNaturalist data

iNaturalist data have seen a tenfold increase in use in scientific literature between 2017 and 2022 (Mason et al. 2025), illustrating the vast potential of these data for use in biodiversity research and conservation. Nevertheless, as described above, these data are complex, heterogeneous along multiple dimensions, and therefore require careful treatment in analyses. The challenge moving forward is not whether iNaturalist data *can* be used, but *how best to use them* and maximize the value of these data for biodiversity and conservation research while avoiding erroneous interpretations. Here, we highlight four different forward-looking uses of iNaturalist data, which we feel represent the next wave of potential research questions.

Secondary data

Although iNaturalist data usage has primarily been focused on the spatiotemporal coordinates of observations, an increasing number of papers extract ecological information from images (Mason et al. 2025). Such “secondary data” (Pernat et al. 2024a), defined as additional information unintentionally captured in species records, are becoming an important and expanding research frontier. Photos from iNaturalist have been used to investigate plant-pollinator interactions (e.g., Pernat et al. 2024b), obtain color data of flowers (Luong et al. 2023), quantify morphological traits such as plumage patterns in birds (Vretos et al. 2021) or color polymorphisms in salamanders (Hantak et al. 2022), study dietary patterns in birds (Panter et al. 2024), identify mating behavior (Ament and Brown 2025), and assess habitat composition (Bolt et al. 2022), as just a few examples. Another example within the frontier of secondary data is the field of phenology, where image timestamps allow flowering times and leaf-outs (Dinnage et al. 2025) as well as animal diel activity patterns (Idec et al. 2025) to be quantified at large geographic and taxonomic scales. This growing field of secondary data use will continue to shape how iNaturalist data are used in the future. As large-scale text-to-image retrieval and multimodal machine learning systems continue to advance (Vendrow et al. 2024), we expect secondary-data applications to play a central role in the next wave of biodiversity research that uses iNaturalist.

Data integration

As researchers extract increasingly diverse ecological information from iNaturalist records, a natural next step is to combine these rich, heterogeneous data with other biodiversity datasets. An increasingly important research frontier is incorporating iNaturalist data with standardized data (e.g., structured or semi-structured survey data or specimen data from natural history collections) to better inform ecological and biogeographic studies (e.g., Yoder et al. 2024), taxonomic

works (e.g., Smith et al. 2022), or conservation and management initiatives (e.g., Ackerfield et al. 2024). In many cases, data integration maximizes the temporal and spatial breadth of the data and improves our understanding of biodiversity from local to global scales (Mesaglio 2025). Although the number and diversity of research papers using integrated datasets have rapidly increased over the last decade (Mason et al. 2025), there is still broad scope and exciting opportunities to further integrate datasets.

Understanding the best approaches and methods for integrating iNaturalist data and traditional data sources, and when or where these approaches are applicable, is critically important. Here, data integration refers not simply to combining datasets, but to analytical frameworks that explicitly account for the differing sampling designs, biases, and uncertainties associated with multiple data sources (Isaac et al. 2020, Mostert and O'Hara 2023). This can be done at both the scale of entire datasets and through more tailored approaches involving individual records or curated partial datasets. These principles may also apply within iNaturalist itself. For example, observations originating from structured projects, coordinated participation events, or different contributor groups may arise from distinct sampling processes that could warrant separate consideration depending on the research question. A key frontier is understanding how integrative approaches can mitigate the idiosyncratic biases present in each data stream. By combining datasets with different strengths and weaknesses, researchers may be able to reduce bias, increase completeness, and better capture true ecological and biogeographic patterns (see Calvente et al. 2023 for a compelling recent example). Developing curated frameworks for combining iNaturalist data with structured surveys, museum collections, and emerging sensor-based datasets will be essential for fully realizing the scientific potential of biodiversity data integration and unlocking the full potential of iNaturalist data.

Estimating non-detections

Even with integrated datasets, many analyses require non-detection information, which has fueled a growing body of research focused on estimating pseudo-absences or non-detections from unstructured biodiversity data (Rapacciuolo et al. 2021, Boyd et al. 2023, Davis et al. 2023, Morera-Pujol et al. 2023, Bowler et al. 2025, van Strien et al. 2026). Because iNaturalist is inherently a presence-only dataset, some analyses and inferences are limited (e.g., Guillera-Arroita et al. 2015), and it means estimating detectability is more difficult than with presence-absence datasets. A promising frontier of current and future research is the development of approaches that derive and/or estimate non-detection information from unstructured iNaturalist data. One direction is the construction of participant-level sampling events, where observations within a place-time window are grouped into a "visit" or "checklist." Rapacciuolo and colleagues (2021) proposed a framework for deriving participant events (akin to checklists in eBird), which could be used to demarcate everything a participant could have documented in a sampling event. For example, when participants blacklight to attract moths, inferences could be made about the species that were not observed during that blacklighting event. Another strategy leverages trait-based or ecological similarity to infer plausible pseudo-absences (e.g., Koen et al. 2025, Sibiya et al. 2025). Species with comparable detectability, body size, microhabitat use, or phenology can be used to estimate when a focal species might rea-

sonably have been observed but was not, thereby providing estimates of non-detections. Best practices for deriving non-detections from unstructured data remain an open challenge (Davis et al. 2023), and future research should focus on clarifying when different strategies are appropriate and how they influence downstream inference.

Trend detection

Estimating biodiversity trends, either at a community level, or a species-specific level, remains an important goal for biodiversity monitoring and research. iNaturalist was launched in 2008, and with increasing time, there is an increasing amount of data to potentially estimate trends. A key frontier for iNaturalist research, therefore, is determining if, when, and how these data can be used to infer occurrence or abundance trends. A fundamental research question is whether the frequency of iNaturalist records in a region approximates the relative abundance of the species in that region. If a stable, interpretable relationship exists, then trend detection using iNaturalist data may become statistically feasible. For instance, if a species is consistently observed hundreds of times per year and then suddenly disappears from new contributions to the platform, would this be driven by a true biological signal, a change in observer behavior, or a consequence of sampling heterogeneity? Disentangling these potential relationships and determining how reliably trends can be identified across space and time are critical open questions for the use of iNaturalist data in biodiversity monitoring.

Emerging research is starting to address these questions. For example, Kaiser and colleagues (2026) used a synthetic-control framework to tease apart causal patterns in iNaturalist reporting, showing how a disturbance event influenced bee detections while appropriately accounting for uneven spatial sampling across regions. Additionally, researchers aiming to make use of iNaturalist data to characterize variation in species' relative abundance, and conduct inference on the drivers of said variation have introduced approaches that attempt to address the problem of spatial variation in effort through model-based inference (Soberón and Christén 2026). Some approaches assume that the reporting rate of a species on iNaturalist is a function of its relative abundance (Ringim et al. 2026), which is reasonable in many cases but breaks down if the abundance of all target species strongly covaries or if the target species makes up a large percentage of all reports in an area of interest.

To fully disentangle biological trends from sampling, research needs to continue to (1) quantify and understand observer effort and reporting behavior, (2) integrate iNaturalist data with structured survey data, and (3) develop and refine statistical approaches suitable for unstructured presence-only time series data for both occupancy and especially relative abundance (Rapacciuolo et al. 2021). Importantly, progress in trend detection will likely depend on developments across several of the other frontiers outlined above.

Conclusions

iNaturalist has become one of the world's most important sources of biodiversity information, and its data are sometimes held to a higher standard than professionally collected datasets despite both arising from imperfect sampling processes (Binley and Bennett 2023). Their scientific value depends on researchers understanding how the data are generated, accessed, filtered, analyzed, and attributed. Through-

out this guide, we highlight best practices that maximize reproducibility, ensure appropriate interpretation of observations, and match analytical approaches to the strengths and limitations of this dynamic dataset. As with any observational dataset, the responsibility for appropriate analysis lies with the researcher. Researchers should explicitly consider the assumptions, biases, and limitations inherent in iNaturalist and ensure that their scientific inferences are appropriate to those characteristics.

Reciprocity and community stewardship

iNaturalist should not be viewed only as a data source: this best-practices-oriented review also focuses on its roots as an evidence-oriented community participatory platform. The scientific value of iNaturalist is built on the enthusiasm of millions of people who choose to share their time, curiosity, and observations. It is essential that researchers approach iNaturalist as a collaborative knowledge-making and sharing system that also supports the betterment of biodiversity research. Respecting licenses and contributor intent is the minimum standard for any use. Wherever possible, researchers should also work to give back to the community. Researchers can credit observers and identifiers in publications, offer co-authorship to significant contributors, share project results in accessible forms, acknowledge taxonomic expertise, and participate in the platform by identifying, annotating, or engaging with users (Callaghan et al. 2022). Such acts of reciprocal contribution strengthen trust, improve data quality, sustain long-term participation, and grow the community. As biodiversity research continues to leverage participatory science data, such principles will help ensure the community builds the richest possible network of data, people, and evidence to support biodiversity research.

Example code and workflows

To help facilitate broad use of iNaturalist data, we have made example code available at <https://github.com/Global-Ecology-Research-Group/Guidelines-and-Best-Practices-for-iNaturalist-Data>. This repository currently contains example R and Python scripts for common ways to access and work with iNaturalist data:

1. **Get Observations:** Code to get observations from a URL request. For example, all butterfly observations in Texas.
2. **Species Counts:** Code to get a table of species counts from one URL request. For example, get a count of observations of all angiosperm species observed in Florida.
3. **Download Images from URL:** Code to download iNaturalist images from a URL path. In the example, observation data are gathered from the iNaturalist API. Then parallel processing is used to download images hosted on Amazon Web Services. Images are named according to observation ID and photo number, so they can be linked to the full observation details later.
4. **Annotation Data:** Code to get annotation data from an API request.
5. **Identifier Data:** Code to get information on identifiers from an API request.
6. **Observer Data:** Code to get information on observers from an API request.
7. **Popular Field Values:** Code to get relevant controlled term values and monthly histogram of observations.

Author contributions

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Supplementary data

Supplementary data are available at *BIOSCI* online.

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Supplementary Text: Guidelines and best practices for the scientific use of global iNaturalist data

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S1: Additional details on iNaturalist data

Metadata details

The date and time of an observation are recorded in the time zone where the observation was recorded. While the date field is required for an observation to be deemed Research Grade, the time field is not; therefore, some observations do not include a time stamp. Additionally, GBIF data (via the Darwin Core Archive) or the data from the iNaturalist's Export Tool provides the full ancestry of the community taxon, including kingdom, phylum, class, order, family, and genus. Geographic coordinates are reported in World Geodetic System 1984 (WGS84), with decimal latitude and longitude provided in separate fields, optionally with an estimate of location uncertainty in meters (see Supplemental Text S2).

Taxon geoprivacy

Taxon geoprivacy obscuration is implemented on an entire taxon basis (usually implemented at the level of species or infraspecies), either globally or within specified regions (e.g., a country or state), and is intended to help combat threats faced by sensitive-location taxa, that could be exacerbated by public data on the animals' occurrence such as habitat disturbance and poaching. Automatic obscuration can be applied to or removed from taxa by site curators when needed and is usually done with reference to sources such as the IUCN Red List, NatureServe, SANBI, or other national or regional institutions that assess the status of species (see list of obscured and private species on iNaturalist [here](#)).

Representation of obscured observations on iNaturalist and in GBIF data

On iNaturalist, when you are viewing an obscured observation, there will be a blue rectangle indicating the region of obscuration and a point which indicates the randomized location. When viewing the map of observations on the Explore page of iNaturalist, transparent points indicate obscured coordinates. Obscured records are made available in GBIF, but the coordinates are the randomized location in the $0.2^{\circ} \times 0.2^{\circ}$ grid cell. With this observation, the accuracy value will encompass the entire obscuration grid, unless the user submitted accuracy is larger, in which case this value will be used. Within GBIF, the Darwin Core field “informationWithheld” is populated with text explaining that the coordinate uncertainty has been increased. On iNaturalist, the date stamp is also obscured to only show month and year for the observation and any identifications that have been added to it; however, the exact observation date and time can be accessed via the iNaturalist API or through GBIF.

Full identification history

While most researchers will use the Community Taxon, the full identification history may provide more fine-scale information about identification trends. This data may be useful to determine identification effort towards specific projects, the top identifiers of a specific taxon, or identification biases. Identification history has been used to assess trends in iNaturalist identification (Campbell et al. 2023), which allowed specific recommendations to be made about iNaturalist data quality. Additionally, summarized user activity trends—including total observations, number of species reported, and number of identifications—can be obtained from iNaturalist. Such data can be useful for assessing engagement trends, identifying top contributors for a specific taxon, or potentially inferring observation effort. For example, researchers have used the iNaturalist API to obtain the dates on which users were active and the number of

observations made on those dates to study observer behavior and iNaturalist data quality (Dimson and Gillespie 2023).

S2: How positional accuracy is determined in iNaturalist

For any given iNaturalist observation, the positional accuracy of its coordinates may have been applied using automated and manual approaches in one of several ways.

1) User A takes a photograph with their smartphone, with location services turned on. A positional accuracy value is embedded in the metadata of the photograph, and when the user uploads this photograph as an observation to iNaturalist the value is automatically applied to the coordinates/location. In this scenario, this value will often be relatively small and accurate, especially in areas with reliable satellite access; however, large errors may occur in some situations, such as where GPS signals are blocked or obstructed (e.g., within a cave or amongst many tall buildings) or if there is a poor Wi-Fi or data connection. This may also happen when photos are taken immediately after opening a phone that hasn't been used recently in the current location, such as when someone is hiking down a trail (in this case, the point may end up centered by the nearest cell tower).

2) User B takes a photograph with their camera which does not have geotagging enabled. When they upload the photograph to iNaturalist, it has no embedded location information at all. The user manually enters coordinates by navigating on the map to a relatively precise location where they took the photograph, and then also manually assigns a positional accuracy value based on their judgement. Here, the positional accuracy is influenced by factors such as the reliability of the user's memory or whether they took notes about the photographic location.

3) User C takes a photograph with their camera which does not have geotagging enabled, however, they have a GPS device running during their walk. When they get home, the user downloads the GPS data as a GPX file, and then uses software on their computer to embed

location data in their photograph by matching the timestamps between the GPX file and the photograph. The coordinates are embedded in the photograph, however, a positional accuracy value is not. When uploading the photograph to iNaturalist, the user follows a similar process to User B, i.e., manually assigning a positional accuracy value, but in this case based on their judgement of how accurate their GPS device was.

4) User D takes a photograph with their camera which does not have geotagging enabled. When they upload the photograph to iNaturalist, it has no embedded location information at all. The user searches for a location using the Google Maps search interface, such as a city or town name (e.g., 'New York') or the name of a reserve or park (e.g., 'Royal National Park'). Google Maps then automatically assigns a pre-determined set of coordinates representing that locality (usually a central point) and a pre-determined positional accuracy value that encompasses the entire locality. Here, the positional accuracy value is generally determined by the size of the feature searched for, i.e., a small rural town will likely have lower level of positional uncertainty, as determined by Google Maps, than an enormous national park or nature reserve.

5) User E takes a photograph with their smartphone, with location services turned on. A positional accuracy value is embedded in the metadata of the photograph, and when the user uploads this photograph as an observation to iNaturalist the value is automatically applied to the coordinates/location. In this scenario, the organism that was photographed is an endangered species threatened by poaching, and so its location data is automatically obscured upon upload. The original positional accuracy is still stored within the record and can be viewed by the observer or other users that they 'trust' with their hidden coordinates, but the public positional accuracy is automatically changed to the diagonal of a $0.2^\circ \times 0.2^\circ$ grid cell.

S3: Observation fields and tags

Unlike the standardized annotation data, observation fields are generated by the iNaturalist community and are often associated with specific projects that seek to gather additional metadata. These observation fields can include free text entry or drop-down options and can be standardized within a project, if the project administrator chooses to do so. Examples of observation fields are species-interaction data (“predated by”, “visited flower of”), behavior observations (“mating”, “nesting”, “pollinating”), and genetic mutations (“albino”, “melanistic”). Because observation fields are user-generated, there may be multiple fields that record essentially the same data (e.g., “habitat”, “habitat type”, “general habitat”, etc.). Tags are user-generated keywords that can be added to any observation by the observer. They allow observations to be searched on iNaturalist, but like observation fields, they are not standardized. Tags are less widely used than observation fields since they are only available via the website interface and can only be added by the observer. Annotations can be accessed from GBIF via columns such as sex, lifeStage, reproductiveCondition, vitality, and establishmentMeans. Observation fields and tags, however, are only available through direct export from iNaturalist or the iNaturalist API.

S4: Retention of Darwin Core data download from GBIF

Darwin Core data downloads from GBIF are retained for at least six months, with cited downloads prioritized for long-term retention. Each GBIF download is assigned a unique DOI that permanently identifies the specific snapshot of data generated by a particular query. However, if GBIF does not detect published literature citing the DOI within six months of export, the dataset records may be deleted, and reproducibility of uncited downloads is not guaranteed. However, the metadata about the export (the DOI, the full query used, the date and time, the number of records, etc.) will be retained in perpetuity.

S5: How to access hidden coordinates

If you are a researcher who needs access to sensitive species locations, the recommended approach is to create a project and allow users to trust the project with hidden coordinates. You can then invite users to join the project and have them grant permission for researchers to access their hidden coordinates. Keep in mind that you are asking users to share sensitive information and you should follow the [Best practices when asking a user for a hidden location](#).

If you are a project curator and users have trusted the project with their coordinates, you can download those coordinates using the iNaturalist Export Tool while logged into your account. In the exported dataset, the columns `private_latitude` and `private_longitude` contain the trusted coordinates. iNaturalist Network members who are trusted with hidden coordinates are trained on how to access hidden coordinates and are required to follow data licensing agreements. If granted access to these hidden coordinates, take special care to avoid sharing this sensitive data in publications or data releases.

S6: How to batch query data

This reference explains how to run multiple queries using the iNaturalist Export Tool and the iNaturalist API when the amount of data needed exceeds the request limits. These request limits are 200,000 observations for the iNaturalist Export Tool and 10,000 observations for the iNaturalist API. The processes for batch querying each dataset rely on the same fundamental steps, but they are executed differently. We first describe the best practices for batch queries, then outline the procedures for (1) the iNaturalist Export Tool and (2) the iNaturalist API.

Best practices for batch queries

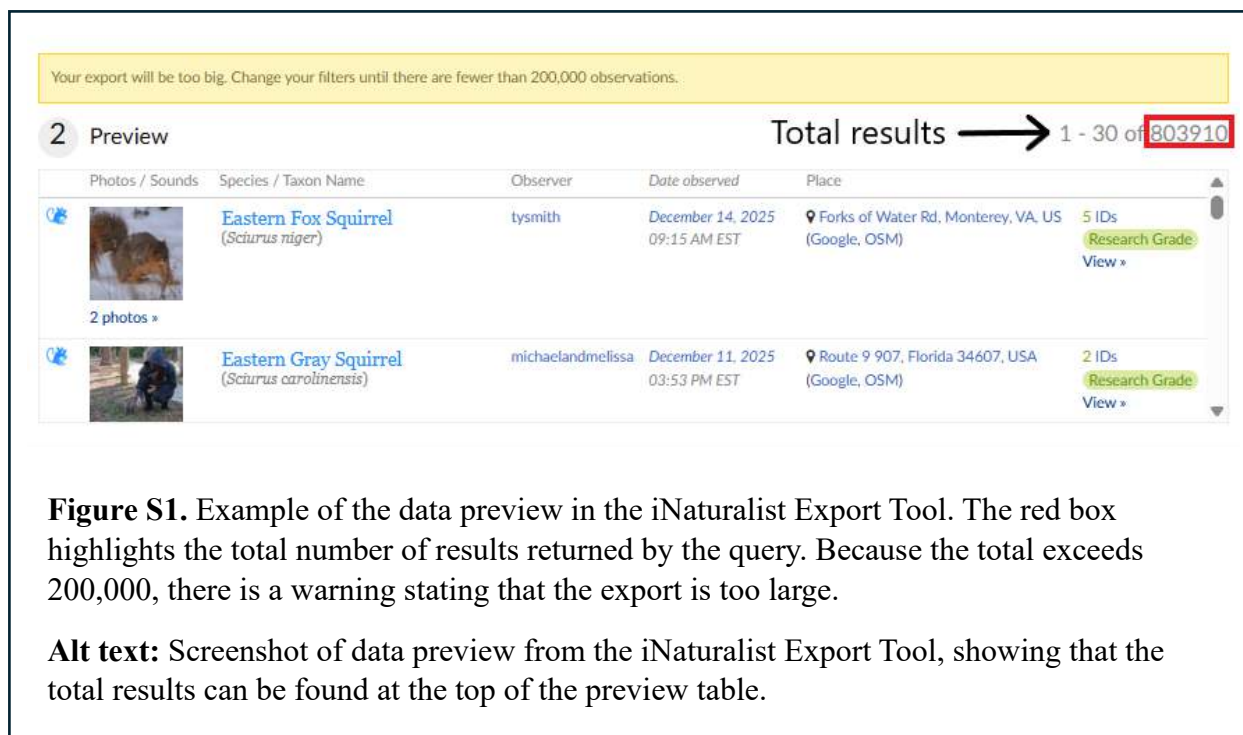
Batch processing data can be time-consuming and places additional strain on servers, so researchers should carefully consider whether it is necessary before running batch queries. If the required data are available through GBIF, they should be downloaded directly from [GBIF](#) (see [iNaturalist Research-grade Observations](#)). If iNaturalist exported data are needed, all possible filters should be applied before downloading. Researchers should also consider whether summarized datasets can be used instead of full observation records. For instance, if only species counts are needed, the `observations/species_counts` API endpoint should be used rather than downloading all observation data. Because the iNaturalist Export Tool allows larger batches to be downloaded (200,000 at once compared to the smaller batches required from the iNaturalist API) the iNaturalist Export Tool should be used when possible. Keep in mind that batch querying should not be used to scrape data. This approach should be reserved for medium-sized datasets and should always follow the [API Recommended Practices](#). If a particular research question requires large amounts of iNaturalist export data, contact help@inaturalist.org for support.

When querying the dataset, you should maximize the number of records downloaded per request. When using the iNaturalist Export Tool, downloads should be close to, but not exceed, 200,000 records. API requests should use the maximum `per_page` value, which is 200 for the observations endpoint, but may be higher for other endpoints. Requesting a small number of large datasets places less strain on the server than making many small requests. When querying for observations without using a specified end date, the dynamic nature of iNaturalist means that the results set may change during your download. For example, if you started a download for Larinae observations in the United States, there would be many new observations created during the time it takes to export the existing observations. For this reason, it is recommended to use a specified end date, and/or plan for a final re-query for the newest observations with data deduplication procedures. Using the API, you can also query in chronological order (`order_by=created_at&order=asc`), fetching the newest observations last.

Query data on iNaturalist Export Tool

One benefit of the iNaturalist Export Tool is that it provides a preview of the data, including the number of observations returned by the query (figure S1). To begin, enter your filtering criteria in the "Create a Query" section, then scroll down to the "Preview" section. Here, you will see a sample of the data returned by the query as well as the number of observations. If your query returns more than 200,000 results, a warning will appear stating: "Your export will be too big. Change your filters until there are fewer than 200,000 observations."

There are multiple ways to filter the data into subsets. One of the common methods is to filter by date using the start and end date fields. If start date is entered with no end date, all data after the start date will be returned. Conversely, if an end date is entered without a start date, all the data



before that date will be returned. Date ranges can be adjusted iteratively until the query returns close to, but fewer than, 200,000 records. This approach is often effective because it allows precise control over the number of records returned.

Another option is to filter by taxon or place, depending on the needed dataset. Taxa can be filtered at any taxonomic level, from broad groups such as kingdoms (i.e., Plants, Animals, Fungi, etc.) or at more specific levels such as family. Place filters can be applied at multiple spatial scales, including country, state, city, place ID, and others. However, this strategy is only effective if each grouping has large amounts of data that are less than 200,000. If that is not the case, then the date method is likely the best option.

Filters can be applied using the fields provided by iNaturalist or by editing the query string shown at the top of the "Create a Query" section (figure S2). When this string is edited, the corresponding fields below are automatically populated. This approach may be more preferable when running multiple queries with a similar structure. In such cases, copy and editing the string for subsequent queries can save time. Saving these parameter strings also helps document how the data were batch queried for future reference.

Regardless of the method used, it is good practice to first define the full set of data needed and note the total number of observations. Subsequent queries should then be designed so that their parameters collectively account for all observations without overlap.

When using the iNaturalist Export Tool, keep in mind that only one request can be submitted to the iNaturalist data server at a time. The time that requests take to finish depends on available server resources. When the export is complete, the "Finished" column will populate and a download option will appear on the left of the request. At this point, the data should be downloaded and saved with a unique filename indicating that it is part of a batch query (e.g., iNat_Export_Batch_1). Once the request is finished, the next query may be submitted.

1 Create a Query

quality_grade=any&identifications=any&place_id=1&taxon_id=45933

Query string

Date range is a common method to query the data

Alt text: Screenshot of the iNaturalist Export Tool, highlighting the data range box with text that says “Date range is a common method to query the data.” Also highlighted is the query string at the top of the page.

Query data using the iNaturalist API

The iNaturalist API call can return up to 200 observations per request, so querying this data may be pertinent. Similar queries as the iNaturalist Export Tool may be used, but the iNaturalist API offers additional parameters that may be used to query the data.

When using the iNaturalist API, you can extract the content of the request to get the total results.

This can be done through the [iNaturalist API](#) webpage by entering filtering parameters then trying out the request. Under "Response Body," at the top of the text you will see total_results.

This tells you how many results the query returns. You can also enter the full API URL into a web browser address bar, or you can do this programmatically by retrieving the content of an API request and examining the total_results data field. For example, see code snippet below:

```
{r}
library(httr)
library(jsonlite)

# API Request URL
api_request <-
  "https://api.inaturalist.org/v2/observations?place_id=21&taxon_id=4715&per_page=0"

# send HTTP GET request to the iNaturalist API
resp <- GET(api_request)

# extract the body of the HTTP response as plain text and convert to R object
data_parsed <- fromJSON(
  content(resp, as = "text", encoding = "UTF-8")
)

# View the total results
print(data_parsed$total_results)
```

If there are more than 200 results, then multiple requests will need to be made to the API to pull all the data. This can be done using the `per_page` parameter for up to 10,000 results, i.e. 50 pages at 200 per page. If more than 10,000 results are needed, then separate queries will need to be made, for example, using the `id_above` parameter (see [Pagination section of API Recommendation Practices](#) for more information). Alternatively, queries can be divided by date, place, taxon, etc., similar to the Export Tool.

When running batch queries keep in mind the iNaturalist rate limit which is 60 requests per minute or 10,000 total API requests per day (~2.8 hours at maximum speed). Separately, you are also limited to 5 GB of iNaturalist-hosted media per hour and 24 GB per day (images hosted on the Amazon Open Data Sponsorship Program are exempt). Throttles should be added into code to ensure that usage is under these limits.

S7: Accessing data via the API

Acquiring iNaturalist data via the API is the more technically involved method of downloading the data, but it can be used to apply special filters to the data or acquire data not available via GBIF or the iNaturalist export tool. iNaturalist currently has three API versions: [v0](#) (written in Ruby on Rails), [v1](#) (in Node), and [v2](#) (also in Node). The v0 API is considered legacy and should only be used for the few authentication endpoints that are not available in v1 or v2. The version recommended for most purposes is v2, with v1 being phased out (see <https://api.inaturalist.org/v2/docs/>). In general, v2 gives smaller, more customized responses. Please refer to iNaturalist's own documentation in the links provided in Supplemental Text S10 for the most up to date recommendations.

When downloading data via the API, iNaturalist throttles the get observations endpoint at 60 requests per minute. Other endpoints may have higher throttle limits, but for all endpoints, they ask users to keep it to 60 requests per minute or lower. The maximum number of calls per day should not exceed 10,000 requests. By setting the `per_page` parameter to its maximum (usually 200 or 500), queries can be consolidated into fewer calls. In terms of media, downloads should be limited to 5GB of data per hour or 24 GB per day for files hosted directly on iNaturalist (there is no limit for images hosted in the [iNaturalist Open Dataset](#)). Given this, the API can be used to acquire small to medium amounts of data, but it is not meant to be used for bulk download.

There are several methods to download data from the [iNaturalist API](#), but the most common approaches involve using coding languages such as Python or R. Both of these programming languages provide unofficial libraries that follow the [iNaturalist API Recommended Practices](#) and can be used to access the iNaturalist API. In Python, the [pyinaturalist](#) library offers

functions for many commonly used API endpoints including observations, identifications, annotations, observation fields, identifications, media, projects, and users. It also supports access to summarized data such as observation histograms, species counts, life lists, and more. In R, the [spocc](#) library may be used to retrieve observation data from GBIF or the iNaturalist API. There is another library in R called [rinat](#), but this library relies on the v0 API and is therefore not recommended for general use.

While using existing libraries may be easier for some use cases, there may be cases where data must be retrieved directly from the API because no prebuilt functions are available. This can be done using Python and R. In Python, the library [requests](#) can be used to send HTTP requests to the API. When making multiple calls to the API in python, the `time.sleep()` function from the built-in `time` module should be used to avoid exceeding the API usage limits. In R, the [httr](#) and [jsonlite](#) libraries are commonly used. The [httr](#) package contains the `GET()` function which sends a request to the API. The [jsonlite](#) package parses the JSON data, the format used by the iNaturalist API, into an R object. Keep in mind that API responses are nested data structures. If you want to preserve this nested structure, you should save the data in .RDS format using `saveRDS()`. When making multiple calls to the API in R, the `Sys.sleep()` function from base R should be used to avoid exceeding API usage limits.

In this section, we gave a brief overview of using the iNaturalist API to download data. Users planning to work with the API should consult available resources (see S10: List of Resources for Accessing iNaturalist Data via the API below). With this article, we also provide code examples to directly retrieve data from the API using R and Python (see S9: Code Examples below). If

challenges arise, [iNatForum](#) can be searched for similar issues or used to post new questions for members of the community to answer

S8: Introduction to modifying request URLs for the API

To acquire data using the iNaturalist API, a customized URL must be constructed. The base URL for the recommended API version is:

`https://api.inaturalist.org/v2`

From there, select an endpoint to retrieve data. The most commonly used endpoint is observations, but other endpoints include places, projects, taxa, observation_tiles, and more. Many endpoints also offer additional resources or paths to access specific types of data. For example, within the observations endpoint, you can access resources such as histogram, identifiers, observers, species_counts, and others. For example, to retrieve species counts specifically, the URL would be:

`https://api.inaturalist.org/v2/observations/species_counts`

To see a list of all endpoints and available parameters, visit the [iNaturalist API website](#). The site conveniently allows users to enter parameters and automatically generates the corresponding request URL. It also provides a view of the response body and headers, which can help determine whether the desired data is returned before downloading it.

When parameters are applied, a “?” is added to the end of the base URL, and all filtering options are listed afterward. Multiple parameters are separated using “&”. For example:

`https://api.inaturalist.org/v2/observations?native=true&taxon_id=47157&acc_below=1000&order=desc&order_by=created_at`

How to acquire ID values

Many of the filtering fields are straightforward. However, some require numeric ID values, such as `taxon_id`, `place_id`, `project_id`, `site_id`, and others. If only one or a few of these ID's are needed, the easiest method to acquire these values is by using the <https://www.inaturalist.org/> website.

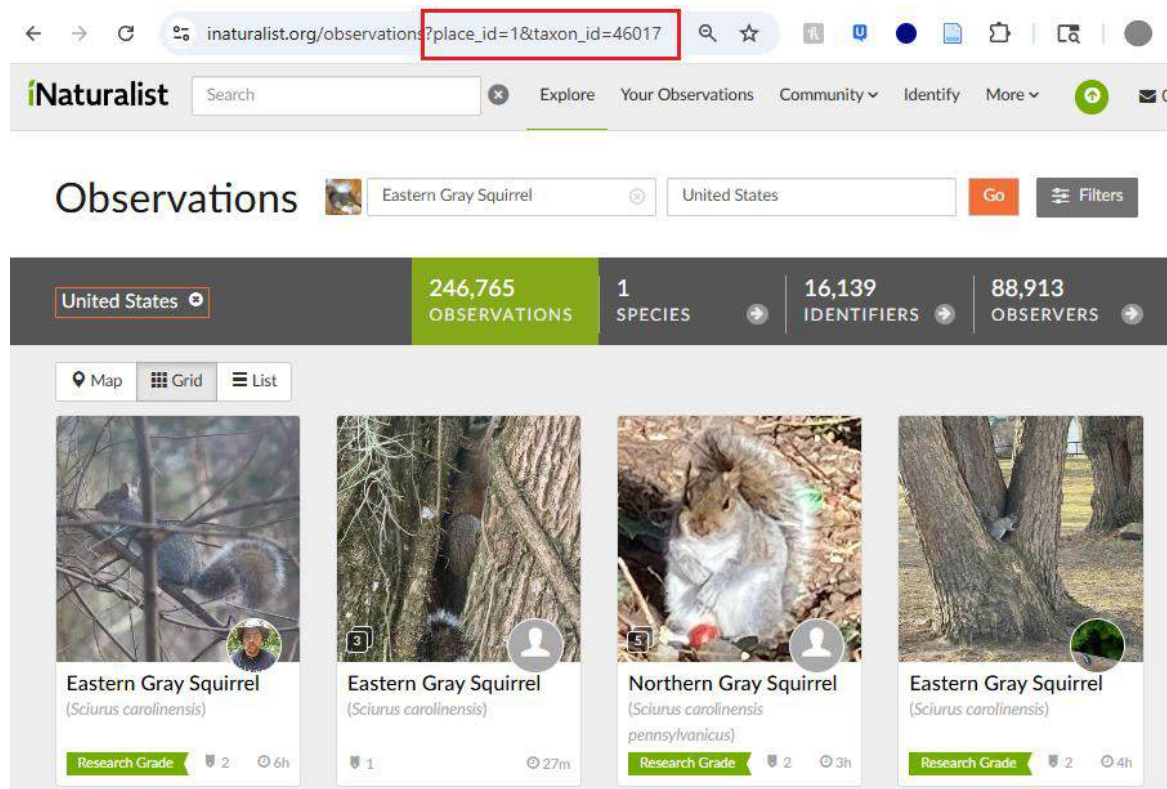


Figure S3. Using the iNaturalist [Explore Page](#) to obtain `taxon_id` and `place_id`. In this example, a search for eastern grey squirrels in the United States was performed, and the URL updated to include the corresponding `taxon_id` and `place_id`. This indicates that the `taxon_id` of eastern grey squirrels is 46017, and the `place_id` of the United States is 1.

Alt text: Screenshot of iNaturalist Explore web page, where the user filtered for eastern grey squirrels in the United States. A red box highlights that the URL now contains a place ID of 1 corresponding to the United States and a taxon ID of 46017 corresponding to eastern grey squirrel.

For `taxon_id`, navigate to the iNaturalist [Explore Page](#) and search for the taxon of interest (figure S3). Once the filter is applied, notice that the URL has changed to show the `taxon_id`. This same methodology can be applied for `place_id`, `project_id`.

How to acquire and handle many ID values

If you need many ID values, downloading summarized datasets can save time. For example, if you need the `taxon_id` for all species in a place, project, from a specific user, or other filtering parameter, the `observations/species_counts` endpoint returns one row per species, including its `taxon_id`. For places, the full list of place IDs and their names and metadata can be downloaded from <http://www.inaturalist.org/places/inaturalist-places.csv.zip>. The `places` endpoint offers two useful options: `places/autocomplete` to search for places beginning with a given text string, and `places/nearby` to find places within a user-defined bounding box.

When setting up a request URL that includes multiple ID values, they should be added as a comma-separated list without spaces. For example:

```
https://api.inaturalist.org/v2/observations?taxon_id=47157,47222,51558,127588,321450
```

Alternatively, the commas can be encoded as “%2C”, which will produce the same result. Keep in mind that multiple IDs for the same parameter act as OR values, not AND values, e.g.

`term_id=1,22` returns observations with either a Life Stage annotation OR an Evidence of Presence annotation.

Resources

[The iNaturalist API documentation](#)

For a more comprehensive overview on how to use iNaturalist's search URLs see [this webpage](#).

S9: Code examples

We provide code examples on GitHub, which can be found at this link:

<https://github.com/Global-Ecology-Research-Group/Example-code-for-accessing-iNaturalist-data>. This repository contains the following scripts in R and Python.

1. **Get Observations:** Code to get observations from a URL request. For example, all butterfly observations in Texas.
2. **Species Counts:** Code to get a table of species counts from one URL request. For example, get a count of observations of all angiosperm species observed in Florida.
3. **Download Images from URL:** Code to download iNaturalist images from URL path. In the example, observation data are gathered from the iNaturalist API. Then parallel processing is used to download images hosted on Amazon Web Services. Images are named according to observation ID and photo number, so they can be linked to the full observation details later.
4. **Annotation Data:** Code to get annotation data from an API request.
5. **Identifier Data:** Code to get information on identifiers from an API request.
6. **Observer Data:** Code to get information on observers from an API request.
7. **Popular Field Values:** Code to get relevant controlled term values and monthly histogram of observations.

S10: List of resources for accessing iNaturalist data via the API

There are many available resources for accessing data via the API. Here we list some of the most relevant resources, categorized by their purpose.

API references

[API recommended practices](#): This webpage documents the best practices for using the iNaturalist API including authentication, pagination, bulk requests, query rate, and building software using the API. Anyone who plans to use the API to access data should read this page and respect the outlined practices.

[iNaturalist v2 API](#): Use this webpage to view all available endpoints and parameters in version 2 of the iNaturalist API. It also allows you to enter parameters directly, and will generate the corresponding request URL along with a sample of the response and headers.

[How to use iNaturalist's search URLs](#): Use this webpage to learn more about modifying request URLs.

[Developers page](#): This webpage contains additional information specific to developers.

R Code

[R for beginners](#): Using R coding to access data via the API requires an understanding of R. This resource provides a good introduction to R, but to become more proficient, you should seek out additional online resources or relevant workshops/courses.

[spocc website](#): This website provides detailed information about the spocc package which can be used to download occurrence data from iNaturalist.

[httr package information](#): A PDF file detailing the 'httr' package used to send requests to the API.

[jsonlite package information](#): A PDF file detailing the 'jsonlite' package used to parse JSON data exported from the API to an R friendly format.

Python Code

[User guide to pyinaturalist](#): A comprehensive guide to using the pyinaturalist package to download iNaturalist data.

[User guide to pyinaturalist](#): A comprehensive guide to using the pyinaturalist package to download iNaturalist data.

[Python for beginners](#): An introduction to installing Python and finding learning resources. Using the example scripts assumes basic familiarity with Python.

[Requests: HTTP for Humans](#): Documentation for the Requests library, used in several example scripts to call the iNaturalist API over HTTP.

[pandas User Guide](#): Topic-based documentation for pandas. The example scripts use pandas to load, reshape, and summarize observation data in tables.

Help Page

[iNatForum](#): Any questions related to iNaturalist—including request URLs or coding questions—can be posted on the iNatForum. Start by searching the forum to see if your question has already been asked and answered. If not, you can create a new post to request help from the iNaturalist community.

S11: Workflows for accessing data summaries, media, and other datasets

In this section, we will discuss how to access data summaries, such as species lists, media, and other datasets using GBIF, the iNaturalist Export Tool, or the iNaturalist API.

Data summaries

Summarized species lists (i.e., checklists of species observed in a particular location), species counts, and spatiotemporal data can also be obtained from multiple sources. Species lists can be downloaded directly from GBIF using the “Species List” dataset option, which provides a list of all species within the specified filters. This dataset includes the number of occurrences for each species, higher taxonomic information (i.e., kingdom, phylum, class, order, family, and genus), and the IUCN Red List Category when available. The iNaturalist API can also be used to obtain species lists or species counts for a specified region, time period, project, or other parameters. This is particularly useful when information not available through GBIF is needed, such as observations of all quality grades (i.e., RG, Needs ID, Casual), or when retrieving species lists for iNaturalist projects or locations that exist only within the iNaturalist platform. Species counts can be accessed through the API endpoint `observations/species_counts`, where users specify the parameters such as the place, time frame, or taxonomic group of interest (Supplemental Text S9). This endpoint reports the number of “leaf taxa,” the default method that iNaturalist uses to count species, in which only the distinct terminal tips of the taxonomic tree are counted to avoid double-counting at different taxonomic ranks. Leaf taxa counts include distinct taxa that have not been identified to the species level—for example, observations identified only to genus when no observations within that genus have been identified to species. As a result, leaf taxa counts are

technically counts of taxa, although iNaturalist uses the term *species counts* because it is more familiar to novice users.

The iNaturalist API can also return summarized temporal data through the observations/histogram endpoint, which provides observation counts binned by a specified interval. Available intervals include month of the year (default), year, month, day, week, hour, and week of the year. These data are useful for visualizing temporal patterns in observations for specified taxa, places, projects, and other filters.

Summarized activity trend data—including total observations, number of species reported, number of identifications for each iNaturalist user, and annotation summary data—are also available from the iNaturalist API. While this information is present in the datasets exported by the iNaturalist’s Export Tool, it is alongside all observations matching the search criteria, which may be more data than required to answer certain research questions. If only activity trends data are needed, the iNaturalist API can be used via the observations/observers and observations/identifiers endpoints. These endpoints provide user information, total observation counts, number of species recorded, activity count, and more (Supplemental Text S9).

Additionally, the observations/popular_field_values endpoint returns annotation summary data, such as the phenology of a species over the course of a year.

Media

Image and sound files are another valuable source of data provided by iNaturalist, which allows multiple images or audio clips to be associated with a single iNaturalist record. Media can be

accessed via the file URL, which can be found in the Darwin Core Archive dataset on GBIF, in the data from the iNaturalist's Export Tool, API responses, or the iNaturalist Open Data Program. For data from GBIF or exported using iNaturalist's Export Tool, only the URL for the first photo or audio recording is provided; whereas the data exported from the iNaturalist API contains URLs for all media associated with an observation record (See Supplemental Text S9). While GBIF only contains open-licensed media, the iNaturalist API can be used to download media of any license, including open licenses not in GBIF (CC-BY-ND, CC-BY-SA, CC-BY-NC-ND, or CC-BY-NC-SA) as well as all rights reserved media. Special attention should be paid to licenses when using this data. In all cases, photos and sounds must be used in compliance with the license terms, and appropriate attribution must be given (See [Creative Commons 2026](#) for details).

Licensed images are stored in the [iNaturalist Open Dataset](#) (iNaturalist 2026a). This dataset uses the Simple Storage Server (S3) from Amazon Web Services (AWS) to provide downloadable images without usage caps. Sound files and all rights reserved images are hosted on iNaturalist servers with a download limit of 5 GB of media per hour or 24 GB of media per day. For AWS hosted images, the URL format is [https://inaturalist-open-data.s3.amazonaws.com/photos/\[photo_id\]/medium.\[file_extension\]](https://inaturalist-open-data.s3.amazonaws.com/photos/[photo_id]/medium.[file_extension]), where photo_id is a unique identifier for each photo that can be linked back to a corresponding observation data record and the file extension is usually jpg or jpeg. iNaturalist-hosted image URLs are of the format [https://static.inaturalist.org/photos/\[photo_id\]/medium.\[file_extension\]](https://static.inaturalist.org/photos/[photo_id]/medium.[file_extension]) and sound URLs are [https://static.inaturalist.org/sounds/\[sound_id\].\[file_extension\]](https://static.inaturalist.org/sounds/[sound_id].[file_extension]). The photo URL can be altered to download different image sizes. In the example above, the photo size is set to “medium” which corresponds to an image size of up to 500 x 500 pixels. Other available size options include “square” (75 x 75 pixels), “thumb” (up to 100 x 100 pixels), “small” (up to 240 x 240 pixels),

“large” (up to 1024 x 1024 pixels), and “original” (up to 2048 x 2048 pixels), although the maximum size never exceeds the original size of the image (e.g., a 400 x 400 pixel image uploaded to iNaturalist will download at that size for the medium, large, and original size options). To save downloading and processing speed and to reduce file storage space, images should be downloaded in the smallest size needed for analysis. Images and sound recordings may be downloaded manually by accessing the URL in a web browser or in an automated process using standard HTTP request methods (e.g., via the base R function `download.file()` or via CURL requests in Python; Supplemental Text S9 and S10). Alternatively, if the photos are to be used for computer vision training, iNaturalist provides a downloadable dataset of 675,000 licensed photos of 5,089 species (iNaturalist Challenge at FGVC: [Van Horn et al. 2017](#)) and a dataset of 3.3 million licensed iNaturalist photos of 10,000 species ([iNaturalist 2021 Challenge at the FGVC8 workshop](#)). However, these datasets were compiled in June 2017 and March 2021, respectively, and are not updated. Researchers seeking to access subsets of observation-associated media for more specific purposes can access them through scripted queries to iNaturalist’s API.

Other datasets

Range maps modeled from iNaturalist data are available under the [iNaturalist Open Range Map Dataset](#) (iNaturalist 2026b), which is updated approximately monthly. The data can be downloaded in bulk by species or taxonomic group. To bulk download range maps, navigate to the Open Range Map Dataset page, then choose the taxonomic group of interest. For some taxonomic groups, there are multiple files to download (i.e., birds, insects, plants). These files are formatted as Geopackages and each contains up to 5,000 range maps. Individual range maps can be downloaded from the Taxon Pages. Individual range maps are not available for all species. To check if one is available for a species of interest, navigate to that species’ about page. Under the

“Map” tab, click the “Overlays” icon. If you see an option for “Expected Nearby Map,” then click the link next to that option. This will take you to the geomodel predictions page for that species (e.g., [iNaturalist 2026c](#)). From here, you can click the “Download Expected Nearby Map as GeoJSON” to get the range map as a GeoJSON file. iNaturalist also provides access to basic heat maps that can be used to visualize observation density (Supplemental Text S12).

Data on other iNaturalist processes, such as identifications, can be obtained using scripted queries of the API. One method is to download all observation data using the base URL <https://api.inaturalist.org/v2/observations>. The json returned from this endpoint can be specified to contain all identification information, including identifier information and whether each identification was categorized as leading (an identification provides more precise taxonomic information within the community taxon), improving (the first identification of an observation that the community agrees with), supporting (an identification that agrees with the community consensus), or maverick (an identification that contradicts community agreement). If observations that were identified by a specified user or list of users are desired, then the `ident_user_id` parameter may be used. Alternatively, if a count of the number of identifications by a user for specified parameters (e.g., specified region, project, taxon, etc.) is desired, then the `observations/identifiers` endpoint can be used. This provides a json with user information, their identification count, and other activity parameters such as number of observations, number of species recorded, number of journal posts, and other information (See Supplemental Text S9).

S12: How to view iNaturalist heat maps

iNaturalist provides access to basic heat maps that can be used to visualize observation density.

Entering the following URL will generate a heat map:

[https://www.inaturalist.org/observations/map?\[insert query\]&color=heatmap](https://www.inaturalist.org/observations/map?[insert query]&color=heatmap)

In place of the string ‘[insert query]’, users can include queries to visualize species based on filters such as place, project, user, year or taxon. For example, a user may want to visualize all iNaturalist observations of peregrine falcons. Navigating to the About page for this species on iNaturalist gives the URL <https://www.inaturalist.org/taxa/4647-Falco-peregrinus>. In this URL, ‘4647’ is the unique identifying number for peregrine falcons. To generate a heat map, a user would therefore use the following URL:

www.inaturalist.org/observations/map?taxon_id=4647&color=heatmap

The heatmap can be zoomed in on, or moved around. Some example queries include:

taxon_id=

place_id=

user_id=

project_id=

year=

Users can also combine multiple queries, separated by the & symbol, in a single URL. For example, the URL

https://www.inaturalist.org/observations/map?taxon_id=4647&place_id=97391&year=2024&color=heatmap

will show a heatmap of observations of peregrine falcons (`taxon_id=4647`) from Europe (`place_id=97391`) from the year 2024 (`year=2024`).

S13: Attribution trends in current literature

To determine the extent to which the current scientific literature provides sufficient information on data attribution to make analyses reproducible, we reviewed 50 articles that relied heavily on iNaturalist data. The articles were randomly selected from the dataset compiled by Mason et al. (2025), which catalogued studies that used iNaturalist data for scientific research. For each article, we assessed whether the authors followed our recommended practices for data attribution. We found that 48% of articles either provided a DOI linking to the GBIF repository containing the occurrence records used in the study or made the iNaturalist data available as supplementary material or through an online repository.

Among the remaining 52% of articles, only one reported the specific query used to access the data. However, 96% described at least some of the filters applied during data retrieval (e.g., taxonomic, geographic, temporal, annotation, observation field, or project-based filters). In contrast, only 23.1% reported the date on which the data were retrieved, making it difficult to reproduce the exact dataset used in the analyses.

Overall, only 40% of articles specified which iNaturalist quality grades (Research Grade, Needs ID, and/or Casual) were included in their analyses. Among studies that accessed data directly through iNaturalist, where users are responsible for complying with observation licensing requirements, only one article explicitly stated that licensing restrictions were followed. Finally, only 12% of articles described how obscured or private locations were handled or whether records were filtered based on coordinate uncertainty.

S14: Suggested reporting template for reproducible use of iNaturalist data

To support transparency, attribution, and reproducibility when using iNaturalist data in scientific analyses, we recommend that researchers explicitly document how the dataset used in their study was obtained, filtered, and processed. This process is important considering iNaturalist observations are “living”. Meaning that observations are not archived or versioned, and so features such as licensing and taxonomic identification may change over time for some observations. Observations may also be deleted by the observer at any time. The following template provides a suggested structure that can be adapted for use in the Methods or Supplementary Materials of a publication.

Example reporting template for GBIF Data Downloads

iNaturalist observation data were obtained on **[DATE]** via **GBIF** using the following DOI: **[GBIF DOI]**. The dataset included observations that met the following criteria: Research Grade, belonging to the taxonomic scope **[taxon names or taxon IDs]**, occurring within the spatial extent **[geographic boundaries, place IDs, bounding box, etc.]**, and recorded during the time period **[date range]**.

Example reporting template for data obtained via the iNaturalist API or iNaturalist Export Tool

iNaturalist observation data were obtained on **[DATE]** via **[iNaturalist API / iNaturalist Export Tool / combination of sources]**. The **[iNaturalist API / iNaturalist Export Tool]** was queried using the following parameters: **[list of parameters or provide request / query URL]**.

The dataset included observations that met the following criteria: **quality grade [Research Grade / Needs ID / Casual / combination]**, belonging to the taxonomic scope **[taxon names or**

taxon IDs], occurring within the spatial extent [**geographic boundaries, place IDs, bounding box, etc.**], and recorded during the time period [**date range**]. Additional filters were applied as needed to match the research objective, including [**e.g., annotations, observation fields, project membership, or user-level filters**].

Example reporting template to describe data filtering methods

Licensing filters were applied to ensure compliance with contributor licensing. Observations included licenses [**CC0 / CC-BY / CC-BY-NC / etc.**], and associated media were used only where licenses permitted reuse. All media and observation records were attributed according to the requirements of the corresponding Creative Commons licenses. Geographic uncertainty was handled as follows: observations with coordinate uncertainty greater than [**threshold in meters**] were [**excluded / retained / treated using buffers**]. Observations with obscured or private coordinates were [**excluded / included with randomized coordinates / accessed via trusted coordinates where permitted**]. Taxonomic interpretation followed [**iNaturalist taxonomy / GBIF Backbone Taxonomy / other authority**], accessed on [**DATE**]. Taxonomic updates occurring after this date were not incorporated into the analysis. The processed dataset used for analysis, including observation ids and derived variables, has been archived at [**repository, e.g., Zenodo, Dryad, GitHub**], with the archive DOI [**DOI**]. Scripts used to retrieve and process the data are provided in [**repository link**].

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